A-A-55A
July 28, 1981
SUPERSEDING
A-A-55
September 26, 1979

COMMERCIAL ITEM DESCRIPTION

LANTERN, ELECTRIC
(FLOATING, CONCENTRATED BEAM, WITH WARMING FLASHER)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification W-L-200.

This commercial item description covers watertight, floating, concentrated beam, dry cell operated electric hand lanterns with a red warning flasher.

Salient characteristics:

The lantern shall be a complete unit including battery case, light sources, switch handle, reflector assembly, red warning flasher assembly, necessary wiring and contacts and spare beam lamp. A positive means of holding the 4Fd 6 volt lantern battery in the battery case shall be used. No special tools shall be required to remove or replace the lamps or battery. A metal stand is optional.

Warning flasher. The red warning flasher shall have a minimum of 30 flashes per minute at 4.9 V.

Impact resistance. After being subjected to a storage temperature of $0^{\circ} \pm 2^{\circ}$ F for 2 hours, the plastic lantern shall show no evidence of damage when subjected to an impact of 1 foot-pound, using a 1 pound steel ball. The impact shall be applied to the base (side opposite the handle), the end opposite the reflector, and the lens ring.

Durability drop. After the plastic lantern has been at a temperature of $75^{\circ} \pm 5^{\circ}$ F for 2 hours, the lantern (batteries included) shall show no evidence of damage when dropped from a minimum height of 3 feet into a box of loose dry sand with a minimum depth of 1 foot. Drop the lantern 3 times on the base and 3 times on the end opposite the reflector. Battery damage shall not be cause for rejection. The lantern shall be operable after this test. The switch may be covered with tape to exclude the sand.

Switch. The switch shall have at least 4 positions including "off" "on" for concentrated beam, "on" for warning flasher, and "on" for both. The switch shall remain electrically operative after 25,000 cycles from "off" through all "on" positions and back to "off". The battery shall be used in the switch test.

Water resistance. No water shall enter the lantern (with battery) during submergence for 30 minutes in a 3 percent NaCl solution at a minimum depth of 3 feet.

Floating. The lantern with battery shall remain at least 1 inch above the surface of a 3 percent NaCl solution after floating for 2 hours.

Concentrated beam. The concentrated light beam shall have a maximum diameter of 10 inches at a distance of 60 inches from the lantern lens.

Beam candlepower. The minimum beam candlepower (candelas) of the concentrated beam shall be 5,000 at 5.2 V. The test distance between lantern and photoreceptor shall be 10 feet. The photometer shall be calibrated using the standard lamp with a color temperature of 2856° K. The photoreceptor shall have a minimum diameter of 1 inch and be equipped with a light filter to yield a response curve matching that or the International Commission on Illumination standard observer. The calculation shall be as follows:

$I = ED^2$

Where: I = Luminous intensity of the light beam in candelas
E = Illuminance reading of photometer calculated in footcandeles
D = Distance between the lens of the lantern and photoreceptor in feet

The lantern shall have no defects that affect serviceability or appearance.

FSC 6230

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Certification: The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided under the provisions of the contract.

Regulatory requirements. The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging, packing, and marking. Packaging, packing, and marking shall be as specified in the contract or order.

Preparing Activity:

GSA-FSS

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